

HAZOP Study - Concentrator (Selected Nodes)

Doc No. RB-4410-STD-HAZOP-001 Rev 1. Recommendations are mandatory unless formally closed.

Node ref	Node	Deviation	Cause	Consequence	Existing safeguard	Recommendation
HZ-31-02	SAG mill hydrostatic lift system	More pressure	HP lift pump dead-heads against closed valve	Oil line rupture, fire at mill	PSV-3105 set <= system design pressure 170 bar(g)	Confirm PSV-3105 set pressure on P&ID matches data sheet
HZ-31-04	SAG mill trunnion bearings	More temperature	Loss of lube oil flow / cooling	Bearing wipe, 10-14 day unplanned outage	TT-3101/3102 TAHH trip via SIF-3101	Align TAHH setpoint between instrument data sheet and cause & effect
HZ-45-03	PAX xanthate mixing room	Other than (decomposition)	Wet/hot PAX, acid contamination	CS2 accumulation - flash fire / toxic exposure	Extraction fan FN-4501	Install CS2 gas detector AT-4501 (alarm 10 % LEL) interlocked to FN-4501 high speed and PAX feeder stop
HZ-45-06	Frother storage TK-4511	Other than (wrong reagent)	Frother specification changed	Hazardous-area classification invalid	Area classified Zone 2 for MIBC	Confirm frother identity on P&ID vs data sheet and SDS
HZ-61-01	Tailings pumping to TSF	More pressure	Blocked line / valve closure at TSF	Pipeline rupture, tailings release to environment	PT-6103 PAHH trip SIF-6101 (SIL 2)	Confirm stage-3 casing design pressure on P&ID

Risk matrix (likelihood x severity 1-5)

	S1	S2	S3	S4	S5
Rare	L	L	M	M	H
Unlikely	L	M	M	H	H
Possible	M	M	H	H	H
Likely	M	H	H	H	H
Almost certain	H	H	H	H	H